Report on "The Effect of Local Road Maintenance Tax Cuts on House Values"

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Overall evaluation

This is an interesting paper on a relevant topic. The authors leverage rich data, combining property sale prices and Google Street View images to measure road quality, and employ a sound identification strategy. I have several comments regarding road quality (measurement and dynamics), methodology, and potential mechanisms (especially compositional effects).

Major comments

1 Road Quality

1. Road quality is measured on a scale from 0 to 2. Is there a specific reason for using only three values? Why not consider expanding to four categories to allow for finer variation?

2. Since road quality is measured both before and after cities vote to cut road maintenance, I suggest plotting the time effects to assess whether the dynamic pattern in road quality aligns with the dynamic pattern in house prices (e.g., in Figure 8). This would strengthen the plausibility of the link between road quality and property values.

2 Methodology

1. Training vs. validation data: Do you divide the data into training and validation sets, or do you include a test set as well?

2. Clarification of assumptions: The paper references Cellini, Ferreira, and Rothstein (2010), stating that "when the elections are independent, the ITT estimator equals the TOT estimator." This point could benefit from a brief explanation (2–3 lines) or inclusion in a footnote for clarity.

3. Robust methodology: Recent work by Hsu and Shen (2024) on dynamic regression discontinuity may be relevant. Notably, they revisit Cellini et al. (2010). The authors should clarify whether the approach in Hsu and Shen (2024) is applicable in the context of this paper, and whether it would alter any of the conclusions.

4. Exogeneity of the timing of the election assessment: On page 17, there is a discussion about whether economic conditions influence the timing of the election. Is there evidence that cities in worse economic conditions are more/less likely have an election, while those in better conditions are less/more likely to do so? This would raise questions about the exogeneity of the timing of the election.

3 Potential Mechanisms

1. Other types of levies: In Appendix C (Table 12), the authors report regression results linking road tax levy referendum outcomes to other types of levies (e.g., police). This table is not discussed in the main text, but it should be. Given the positive correlation between road and police levies, the possibility that changes in crime rates drive the observed house price changes should be considered. Crime variables are not included in the main analysis; tracking their evolution over time before and after the vote would help identify underlying mechanisms.

2. Compositional effects: A broader concern is whether cuts in public goods provision affect migration patterns, potentially leading to compositional changes in neighborhoods.

The role of compositional effects is discussed in the literature, including experimental work on the impact of paving roads on home values by Gonzalez-Navarro and Quintana-Domeque (2016). This could be an alternative explanation for the observed effects on house prices. It would be valuable to explore dynamic effects on covariates—not only baseline differences (discussed on page 14), but also their evolution over time. Do individuals who value road quality tend to move out when maintenance is reduced? Is the pattern in Figure 8 partly capturing these compositional shifts?

Minor Comments

1. Tables: The format of Table 8 seem more appropriate and complete than that of Tables 4-7, which only report estimates from $t + 4$ onwards.

2. Standardized differences: It may be helpful to report the normalized difference in means (Imbens and Rubin, 2015) in Table 2 for clearer comparison of baseline characteristics:

$$NormDiff = \frac{\overline{X}_T - \overline{X}_C}{\sqrt{(S_T^2 + S_C^2)/2}}$$

where $\overline{X}_T$ (resp. $\overline{X}_C$) is the average of the covariate for the treatment (resp. comparison) group, and S_T^2 (resp. S_C^2) is the sample variance of the covariate for the treatment (resp. comparison) group. See Imbens and Rubin (2015, p. 277) for a general rule of thumb to assess imbalance.

3. Figures 5, 6, and 7: These figures could be improved for better visualization of patterns.

References

- Imbens, G. W., and Rubin, D. B. (2015) "Causal inference for statistics, social, and biomedical sciences: An introduction." Cambridge University Press.

- Gonzalez-Navarro, M., and Quintana-Domeque, C. (2016) "Paving streets for the poor: Experimental analysis of infrastructure effects," *The Review of Economics and Statistics*

98 (2): 254–267.

- Hsu, Y-C., and Shen, S. (2024) “Dynamic regression discontinuity under treatment effect heterogeneity,” *Quantitative Economics* 15 (4): 1035–1064.